

解: $\lambda_1 = 4$ (重根)
Smith 标准形:
$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \lambda(\lambda-1) & 0 & 0 \\ 0 & 0 & \lambda^2(\lambda-1)^3 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$
 特征行列式:
$$\begin{aligned} D_1(\lambda) &= 1 \\ D_2(\lambda) &= \lambda(\lambda-1) \\ D_3(\lambda) &= \lambda^3(\lambda-1)^4 \end{aligned}$$

解:
$$A = \begin{bmatrix} 1 & 3i & 0 \\ -3i & 1 & 0 \\ 0 & 0 & 4 \end{bmatrix} \quad |\lambda I - A| = \begin{vmatrix} \lambda-1 & -3i & 0 \\ 3i & \lambda-1 & 0 \\ 0 & 0 & \lambda-4 \end{vmatrix} = (\lambda-4)^2(\lambda+2)$$

当 $\lambda = 4$ 时
$$4E - A = \begin{bmatrix} 3 & -3i & 0 \\ 3i & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -i & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$
 $\alpha_1 = (1, 1, 0)^T \Rightarrow v_1 = (\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0)^T$
 $\alpha_2 = (0, 0, 1)^T \Rightarrow v_2 = (0, 0, 1)^T$

当 $\lambda = -2$ 时
$$-2E - A = \begin{bmatrix} -3 & -3i & 0 \\ 3i & -3 & 0 \\ 0 & 0 & -6 \end{bmatrix} = \begin{bmatrix} 1 & -i & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$
 $\alpha_3 = (1, 1, 0)^T \Rightarrow v_3 = (\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0)^T$
$$U = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \end{bmatrix} \quad U^H A U = \begin{bmatrix} 4 & & \\ & 4 & \\ & & -2 \end{bmatrix}$$

令 $X = UY$, 其中 $Y = (y_1, y_2, y_3)^T$. 代入二次型得:

$$f(x_1, x_2, x_3) = X^H A X = Y^H (U^H A U) Y = 4\bar{y}_1 y_1 + 4\bar{y}_2 y_2 - 2\bar{y}_3 y_3$$

三. 解: $|\lambda I - AB| = 0 \Rightarrow A$ 正定 H 阵 $\therefore \exists$ 可逆矩阵 $Q: A = Q^H Q$

$$\begin{aligned} 0 &= |\lambda I - AB| = |\lambda I - Q^H Q B| = |\lambda Q^H Q - Q^H Q B| = |Q^H (\lambda I - QB) Q| \\ &= |Q^H| |\lambda I - QBQ^H| |Q| = |\lambda I - QBQ^H| \end{aligned}$$

$\therefore \lambda$ 也是 QBQ^H 的特征值.

又 QBQ^H 为 H 阵 $\therefore \lambda$ 为实数 同理证 BA 的特征值有实部.

四. 解: $A = \begin{bmatrix} 2 & -1 \\ 0 & 0 \\ -2 & 1 \end{bmatrix} \quad A^H = \begin{bmatrix} 2 & 0 & -2 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix} \quad AA^H = \begin{bmatrix} 5 & 0 & -5 \\ 0 & 0 & 0 \\ -5 & 0 & 5 \end{bmatrix} \quad |\lambda I - AA^H| = \lambda^2(\lambda-10)$

$\lambda_1 = 10, \lambda_2 = \lambda_3 = 0 \therefore$ 正交阵 $Q: Q^H = Q^{-1}: \Delta = \sqrt{10}$

当 $\lambda_1 = 10$ 时:
$$10E - AA^H = \begin{bmatrix} 5 & 0 & -5 \\ 0 & 0 & 0 \\ 5 & 0 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \alpha_1 = (1, 0, 1)^T \Rightarrow u_1 = (\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}})^T \quad v_1 = u_1$$

$$v_1 = A^H u_1 \Delta^{-1} = \begin{bmatrix} 2 & 0 & -2 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix} \frac{1}{\sqrt{10}} = \begin{bmatrix} \frac{2}{\sqrt{5}} \\ 0 \\ \frac{2}{\sqrt{5}} \end{bmatrix}$$

$$v_2 = (\frac{-1}{\sqrt{5}}, \frac{2}{\sqrt{5}})^T$$

当 $\lambda_2 = 0$ 时:
$$-AA^H = \begin{bmatrix} -5 & 0 & 5 \\ 0 & 0 & 0 \\ 5 & 0 & -5 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & -1 \end{bmatrix} \quad \alpha_2 = (1, 0, -1)^T \Rightarrow u_2 = (\frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}})^T$$

$$v_2 = A^H u_2 \Delta^{-1} = \begin{bmatrix} 2 & 0 & -2 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ -\frac{1}{\sqrt{2}} \end{bmatrix} \frac{1}{\sqrt{10}} = \begin{bmatrix} \frac{2}{\sqrt{5}} \\ 0 \\ -\frac{2}{\sqrt{5}} \end{bmatrix}$$

$$\text{当 } \lambda_1 = \lambda_2 = 0 \text{ 时}$$

$$-AA^H = \begin{bmatrix} -5 & 0 & i \\ 0 & 0 & 0 \\ 5 & 0 & -i \end{bmatrix} \approx \begin{bmatrix} -1 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \alpha_2 = (1, 0, 1)^T \quad \mu_2 = (\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}})^T$$

$$\alpha_1 = (0, 1, 0)^T \quad \mu_1 = (0, 1, 0)^T$$

$$A = \begin{bmatrix} 1 & 1 & -\frac{1}{\sqrt{2}} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

$$A = BC$$

$$A^+ = C^H(C C^H)^{-1}(C^H B)^+ B^H$$

$$= \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}$$

五、证明：

证法一：~~证法一~~ $\frac{1}{2} \|A\| \neq 0$ 时， $|a_{ij}| > 0 \therefore \|A\| > 0$

$\frac{1}{2} \|A\| = 0$ 时， $|a_{ij}| = 0 \therefore \|A\| = 0$

证法二： $\|kA\| = \sqrt{mn} \max_{i,j} |k a_{ij}| = |k| \sqrt{mn} \max_{i,j} |a_{ij}| = |k| \|A\|$

证法三： $A \in C^{m \times n}, B \in C^{n \times n}$

$$\|A+B\| = \sqrt{mn} \max_{i,j} |a_{ij} + b_{ij}| \leq \sqrt{mn} \max_{i,j} |a_{ij}| + \sqrt{mn} \max_{i,j} |b_{ij}| = \|A\| + \|B\|$$

证法四： $A \in C^{m \times n}, B \in C^{n \times l}$

$$\begin{aligned} \|AB\| &= \sqrt{ml} \max_{i,j} \left| \sum_{k=1}^n a_{ik} b_{kj} \right| \leq \sqrt{ml} \max_{i,j} \sum_{k=1}^n |a_{ik}| |b_{kj}| \leq \sqrt{ml} \cdot n \max_{i,k} |a_{ik}| \max_{k,j} |b_{kj}| \\ &= \sqrt{mn} \cdot \max_{i,k} |a_{ik}| \cdot \sqrt{nl} \max_{k,j} |b_{kj}| = \|A\| \|B\| \end{aligned}$$

$$\lambda E - A = \begin{vmatrix} \lambda & -1 & -i \\ 1 & \lambda & 0 \\ -i & 0 & \lambda \end{vmatrix} = \lambda(\lambda^2 + 1) \quad \lambda_1 = 0 \quad \lambda_2 = i \quad \lambda_3 = -i$$

① $\lambda_1 = 0$ 时

$$-A = \begin{bmatrix} 0 & 1 & i \\ 1 & 0 & 0 \\ -i & 0 & 0 \end{bmatrix} \approx \begin{bmatrix} 0 & 1 & i \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \alpha_1 = (0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})^T$$

② $\lambda_2 = i$ 时

$$iE - A = \begin{bmatrix} i & -1 & -i \\ 1 & i & 0 \\ -i & 0 & i \end{bmatrix} \approx \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \alpha_2 = (1, 0, 1)^T$$

③ $\lambda_3 = -i$ 时

$$-iE - A = \begin{bmatrix} -i & -1 & -i \\ 1 & -i & 0 \\ -1 & 0 & -i \end{bmatrix} \approx \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \alpha_3 = (-1, 0, 1)^T$$

$$\therefore \text{解: } |\lambda E - A| = \begin{vmatrix} \lambda & -1 & -i \\ 1 & \lambda & 0 \\ -i & 0 & \lambda \end{vmatrix} = \lambda(\lambda^2 + 2) = 0$$

$$\lambda_1 = 0 \quad \lambda_2 = \sqrt{2}i \quad \lambda_3 = -\sqrt{2}i \quad \alpha_1^H = (0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})^T$$

$$\textcircled{1} \lambda_1 = 0 \text{ 时 } \alpha_1 = (0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})^T \quad \alpha_1^H = (\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})^T$$

$$\textcircled{2} \lambda_2 = \sqrt{2}i \text{ 时 } \alpha_2 = (\frac{-1}{\sqrt{2}}, \frac{1}{2}, -\frac{1}{2})^T \quad \alpha_2^H = (\frac{1}{\sqrt{2}}, \frac{1}{2}, \frac{1}{2})^T$$

$$\textcircled{3} \lambda_3 = -\sqrt{2}i \text{ 时 } \alpha_3 = (\frac{1}{\sqrt{2}}, \frac{1}{2}, \frac{1}{2})^T \quad \alpha_3^H = (\frac{1}{\sqrt{2}}, \frac{1}{2}, \frac{1}{2})^T$$

$$G_1 = \alpha_1 \alpha_1^H = \begin{bmatrix} 0 & 0 & 0 \\ 0 & \frac{1}{2} & -\frac{1}{2} \\ 0 & -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

$$G_2 = \alpha_2 \alpha_2^H = \begin{bmatrix} \frac{1}{2} & \frac{-1}{2\sqrt{2}} & \frac{1}{2\sqrt{2}} \\ \frac{-1}{2\sqrt{2}} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{2\sqrt{2}} & \frac{1}{4} & \frac{1}{4} \end{bmatrix}$$

$$G_3 = \alpha_3 \alpha_3^H = \begin{bmatrix} \frac{1}{2} & \frac{1}{2\sqrt{2}} & \frac{1}{2\sqrt{2}} \\ \frac{1}{2\sqrt{2}} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{2\sqrt{2}} & \frac{1}{4} & \frac{1}{4} \end{bmatrix}$$

$$A = \sqrt{2}i G_2 - \sqrt{2}i G_3$$

$$\text{E. 解: } |\lambda E - A| = (\lambda - 3)^4 \quad \text{最少次数: } (\lambda - 3)^3 \quad J = \begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 3 & 1 & 0 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

$$\text{令 } p(x) = a_0 + a_1 x + a_2 x^2 \quad p'(x) = a_1 + 2a_2 x \quad p''(x) = 2a_2$$

$$\begin{cases} f(3) = p(3) = a_0 + 3a_1 + 9a_2 \\ f'(3) = p'(3) = a_1 + 6a_2 \\ f''(3) = p''(3) = 2a_2 \end{cases}$$

$$\Rightarrow \begin{cases} a_0 = f(3) - 3f'(3) + \frac{1}{2}f''(3) \\ a_1 = f'(3) - 3f''(3) \\ a_2 = \frac{1}{2}f''(3) \end{cases}$$

$$\Rightarrow f(A) = a_0 E + a_1 A + a_2 A^2$$

$$\textcircled{1} \text{ 令 } f(A) = e^{tA} \quad f'(A) = t e^{tA} \quad f''(A) = t^2 e^{tA}$$

$$f(3) = e^{3t} \quad f'(3) = t e^{3t} \quad f''(3) = t^2 e^{3t}$$

$$f(A) = \begin{bmatrix} f(3) \\ f'(3) - \frac{5}{2}f''(3) \\ f''(3) - \frac{5}{2}f'(3) \\ f(3) \end{bmatrix} = \begin{bmatrix} e^{3t} \\ (t - \frac{5}{2})e^{3t} \\ (t - \frac{5}{2})e^{3t} \\ e^{3t} \end{bmatrix}$$

$$\textcircled{2} \text{ 令 } f(A) = \cos \frac{\pi}{2} A \quad f'(A) = -\frac{\pi}{2} \sin \frac{\pi}{2} A \quad f''(A) = -\frac{\pi^2}{4} \cos \frac{\pi}{2} A$$

$$f(3) = \cos \frac{\pi}{2} \cdot 3 = 0 \quad f'(3) = \frac{\pi}{2} \quad f''(3) = 0$$

$$f(\cos \frac{\pi}{2} A) = \cos \frac{\pi}{2} A = \begin{bmatrix} 0 & 0 & 0 & 0 \\ \frac{\pi}{2} & 0 & 0 & 0 \\ 0 & \frac{\pi}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

18 解: 证明:

$$\cos(2\pi B + A) = \cos 2\pi B \cdot \cos A + \sin 2\pi B \cdot \sin A = E \cdot \cos A = \cos A$$

$$\therefore E = \begin{bmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{bmatrix}_{n \times n} \quad 2\pi B = \begin{bmatrix} 2\pi & & \\ & \ddots & \\ & & 2\pi \end{bmatrix} \quad \cos 2\pi B = \begin{bmatrix} \cos 2\pi & & \\ & \ddots & \\ & & \cos 2\pi \end{bmatrix} = \begin{bmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{bmatrix}$$

$$\text{同理: } \sin 2\pi B = \begin{bmatrix} 0 & & \\ & \ddots & \\ & & 0 \end{bmatrix}$$

2. 证明:

$$\frac{dA(t)}{dt} = \begin{bmatrix} 2e^{2t} & 0 & 0 \\ 0 & 0 & 2t \\ 0 & 2t & 0 \end{bmatrix}.$$

$$\frac{d^2A(t)}{dt^2} = \begin{bmatrix} 4e^{2t} & 0 & 0 \\ 0 & 0 & 2 \\ 0 & 2 & 0 \end{bmatrix}$$

$$\int_0^{x^2} A(t) dt = \begin{bmatrix} \frac{1}{2}e^{2x^2} - \frac{1}{2} & 0 & 0 \\ 0 & 3x^2 & \frac{1}{3}x^6 \\ 0 & \frac{1}{3}x^6 & 3x^2 \end{bmatrix}.$$

$$\therefore \frac{d}{dx} \left(\int_0^{x^2} A(t) dt \right) = \begin{bmatrix} \frac{e^{2x^2}}{2x} & 0 & 0 \\ 0 & 6x & 2x^5 \\ 0 & 2x^5 & 6x \end{bmatrix}$$